

Mitigating the Photo-Sketch Domain Gap: Empirical Analysis of Target Fine-Tuning and Style-Transfer Augmentation in Few-Shot Vehicle Classification

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Abstract—Deep Convolutional Neural Networks (CNNs) exhibit remarkable generalization under the independent and identically distributed (i.i.d.) assumption but often suffer catastrophic performance degradation when exposed to out-of-distribution data. This paper investigates the phenomenon of Domain Shift using the DomainNet benchmark, specifically focusing on the Source-to-Target transition from real photographs (*Real*) to freehand drawings (*Sketch*) for a 6-class vehicle classification task. Operating under a strict few-shot constraint of 50 training images per class, we evaluate a pretrained ResNet-50 baseline against two domain adaptation strategies: Target Fine-Tuning and Style-Transfer Augmentation via an L-BFGS optimized Neural Style Transfer (NST) pipeline. Our empirical evaluation demonstrates that the baseline model achieves 93.65% accuracy on the source domain but drops to 55.45% on the target domain, yielding a severe domain shift penalty ($\Delta_{shift} = 0.3819$). Target Fine-Tuning drastically mitigates this gap ($\Delta_{shift} = 0.1194$), improving target accuracy to 81.35%. Furthermore, unsupervised Style-Transfer Augmentation improves target accuracy to 64.86% without requiring target-domain labels. We support our quantitative findings with t-SNE feature projections and Grad-CAM attention maps, providing a deep discussion on the mechanisms of feature distortion and textural reliance in modern CNNs.

Index Terms—Domain Shift, Few-Shot Learning, Neural Style Transfer, Domain Adaptation, Convolutional Neural Networks, DomainNet.

I. INTRODUCTION

Transfer learning has fundamentally transformed computer vision, enabling the reuse of deep hierarchical representations optimized on massive datasets, such as ImageNet, for downstream tasks with scarce labeled data [1], [2]. However, while feature extraction pipelines excel when source and target domains share similar visual distributions, they frequently fail when evaluated across different visual modalities. This degradation is formalized as *Domain Shift* [4].

This study addresses Challenge 7 (Group 2) assigned by the Systems Engineering Program at Universidad Distrital Francisco José de Caldas. The task involves a 6-class high-level semantic identification workflow comprising vehicles: *airplane*, *bicycle*, *bus*, *car*, *motorcycle*, and *train*. The computational challenge requires extracting abstract representations from high-fidelity photographs (*Real* domain) and enforcing a robust generalization mapping onto abstract, edge-only representations without color or texture (*Sketch* domain). This is

strictly bounded by a few-shot limitation of 50 training tokens per category.

To reconcile this structural disconnect, we implement and analyze a sequence of deep network adaptations:

- A baseline formulation using Feature Extraction and Fine-Tuning.
- Generative data interpolation using Gatys’ Neural Style Transfer (NST) paradigm [5].
- A Target Fine-Tuning strategy utilizing sparse labeled target data.
- An unsupervised Style-Transfer Augmentation strategy.

II. RELATED WORK

A. Deep Transfer Learning

Transfer learning leverages prior knowledge to solve target tasks efficiently. Early layers in Convolutional Neural Networks (CNNs) learn universal Gabor-like edge detectors, while deeper layers capture task-specific semantics [2]. Fine-tuning a pretrained ResNet-50 [1] remains the standard approach for few-shot image classification, significantly outperforming networks trained from scratch [3].

B. Domain Adaptation and DomainNet

Unsupervised Domain Adaptation (UDA) seeks to align feature distributions between a labeled source domain and an unlabeled target domain. Peng et al. [4] introduced the DomainNet dataset and the Moment Matching for Multi-Source Domain Adaptation (M3SDA) framework to address multi-domain disparities. Other adversarial approaches, such as Domain-Adversarial Neural Networks (DANN) [7], utilize gradient reversal layers to force feature extractors to become domain-invariant.

C. Neural Style Transfer (NST)

Gatys et al. [5] demonstrated that CNNs can independently extract content and style representations. By matching Gram matrices of deep feature maps, an image can be synthesized to retain the structural content of one image and the textural style of another. While feed-forward networks [6] offer faster synthesis, L-BFGS pixel-level optimization remains the gold standard for high-fidelity style transfer, functioning here as a powerful tool for synthetic data augmentation.

III. DATA DESCRIPTION AND PREPROCESSING

A. Dataset Sources and Partitions

We utilize the DomainNet benchmark [4], focusing specifically on the *Real* (Source) and *Sketch* (Target) domains. To rigorously adhere to the few-shot constraints, the data was partitioned as follows:

- **Source Train:** Exactly 50 images per class from the *Real* domain.
- **Source Validation:** Exactly 50 images per class from the *Real* domain.
- **Source/Target Test:** The remaining images in their respective domains, preserved solely for final evaluation to prevent data leakage.

A naming inconsistency in the original dataset (where the *motorcycle* class was labeled as *motorbike*) was programmatically mapped during the preprocessing pipeline to ensure systemic consistency.

B. Image Preprocessing

Standard ImageNet statistics were enforced across all domains to prevent artificial covariate shifts. All tensors were normalized using $\mu = [0.485, 0.456, 0.406]$ and $\sigma = [0.229, 0.224, 0.225]$, and resized to 224×224 pixels. During training, random horizontal flipping and color jittering were applied to mitigate overfitting on the sparse 50-shot source data.

IV. METHODOLOGY

A. Phase A: Few-Shot Baseline Formulation

We employed a ResNet-50 backbone pretrained on IMAGENET1K_V2. The adaptation proceeded in two optimization phases:

- 1) **Feature Extraction:** The backbone parameters ($\theta_{backbone}$) were frozen (*requires_grad = False*). The fully connected head was replaced by a 6-way linear projection and optimized using Adam ($lr = 10^{-3}$) for 20 epochs.
- 2) **Deep Fine-Tuning:** The constraint was relaxed on the deepest convolutional blocks (*layer3* and *layer4*). The network was fine-tuned for 30 additional epochs using a conservative learning rate ($lr = 10^{-4}$) to prevent catastrophic forgetting.

The severity of the domain gap was quantified using the penalty metric:

$$\Delta shift = Acc_{source} - Acc_{target} \quad (1)$$

B. Phase B: Generative Style-Transfer

To construct a cross-domain bridge without external annotations, we applied NST to generate synthetic training tokens. For a real photograph (Content, I_c) and a sketch (Style, I_s), the synthesized image I_g is optimized directly in pixel space by minimizing:

$$\mathcal{L}_{total}(I_g) = \alpha \mathcal{L}_{content}(I_c, I_g) + \beta \mathcal{L}_{style}(I_s, I_g) \quad (2)$$

Using a VGG-19 feature extractor, content was mapped from *relu4_2*, while style was extracted via Gram Matrices from *relu1_1* through *relu5_1*. Optimization was driven by L-BFGS over 300 steps with an empirically selected ratio of $\alpha/\beta = 10^{-4}$. We generated 180 synthetic images (30 per class).

C. Phase C: Domain Adaptation

Two distinct strategies were evaluated to reduce the $\Delta shift$:

- **Target Fine-Tuning (Target FT):** The baseline model was un-frozen at *layer4* and *fc*, and fine-tuned for 15 epochs strictly on 50 true labeled samples from the *Sketch* training split.
- **Style-Transfer Augmentation (Synth Aug):** An augmented dataset was constructed by concatenating the 50 *Real* images with the 30 *Synthetic* (stylized) images per class. The baseline model was fine-tuned on this mixed dataset.

V. EXPERIMENTAL RESULTS

All experiments were executed while tracking the optimal validation accuracy. Performance metrics reported directly reflect the evaluation on the full held-out test splits.

A. Qualitative Results: Style-Transfer Synthesis

Before evaluating the domain adaptation strategies, we visually verified the output of the Neural Style Transfer (NST) pipeline. As depicted in Figure 1, the L-BFGS optimization effectively bridges the visual gap by transferring the structural edges of the target sketches onto the semantic layout of the source photographs. These synthetic images served as the foundation for the unsupervised augmentation strategy evaluated below.

B. Quantitative Performance Matrix

Table I summarizes the absolute test accuracies and the calculated Domain Shift Penalty across the three tested configurations.

TABLE I: Cross-Domain Performance and Generalization Metrics

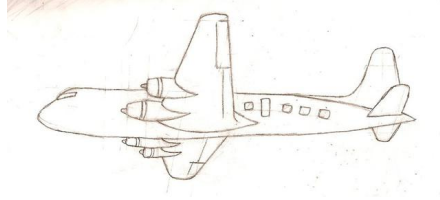
Adaptation Strategy	Acc. Source	Acc. Target	$\Delta shift$
Baseline (Phase A)	0.9365	0.5545	0.3819
Target Fine-Tuning	0.9329	0.8135	0.1194
Style-Transfer Aug.	0.9433	0.6486	0.2946

The **Baseline** model achieved an outstanding Source accuracy of 93.65%. However, upon evaluation on the Sketch dataset, accuracy collapsed to 55.45%. This translates to a severe domain shift penalty of $\Delta shift = 0.3819$.

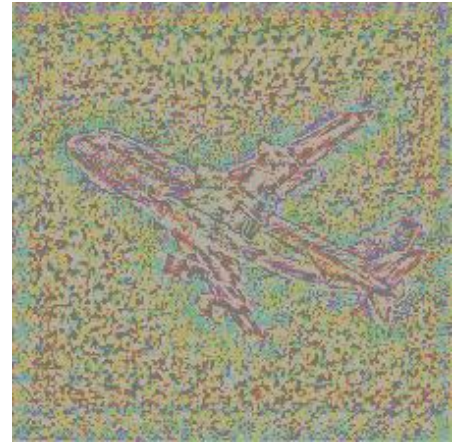
The implementation of **Target Fine-Tuning** dramatically improved the Target accuracy to 81.35%, recovering over 25 percentage points of the lost accuracy, while experiencing a negligible drop in Source performance (93.29%). This strategy compressed the $\Delta shift$ to just 0.1194.



(a) Source Content (I_c): Real



(b) Target Style (I_s): Sketch



(c) Synthesized Image (I_g)



(d) Source Content (I_c): Real



(e) Target Style (I_s): Sketch



(f) Synthesized Image (I_g)

Fig. 1: Qualitative results of the generative Neural Style Transfer (NST) pipeline for synthetic data augmentation. Rows illustrate independent samples from the *airplane* and *car* categories. The L-BFGS optimization perfectly preserves the high-level semantic layout and geometry of the source photograph (a, d) while successfully absorbing the edge-centric, textureless artistic style of the freehand target drawings (b, e) into the final augmented tokens (c, f).

Finally, the **Style-Transfer Augmentation** yielded a Target accuracy of 64.86% (a +9.41% absolute gain over the baseline), while concurrently raising the Source accuracy to a peak of 94.33%. The final domain shift penalty for this unsupervised method was 0.2946.

VI. DISCUSSION

A. Interpretation of Domain Shift Mitigation

The severe 38.19% performance drop observed in the Baseline highlights a fundamental limitation of modern CNNs: a strong inductive bias towards high-frequency textures and local color gradients rather than global object shape. When these textures are stripped away in the *Sketch* domain, the unadapted ResNet-50 fails to find the discriminative features it relied upon during ImageNet and Phase A training.

Target Fine-Tuning proved to be the most potent intervention. By explicitly exposing the deep convolutional filters

(layer4) to actual target-domain contours during backpropagation, the model successfully recalibrated its decision boundaries to prioritize edge geometry over color. However, this method assumes the availability of labeled target data, which is often unfeasible in real-world deployments.

In contrast, Style-Transfer Augmentation acts as a powerful unsupervised regularizer. By injecting synthetic images that maintain the geometric structure of real cars but exhibit the chaotic, colorless texture of sketches, we force the network to become "texture-invariant." Although the +9.41% gain in target accuracy does not match the supervised fine-tuning approach, it validates the hypothesis that generative textural interpolation can partially bridge the domain gap without requiring a single target-domain annotation.

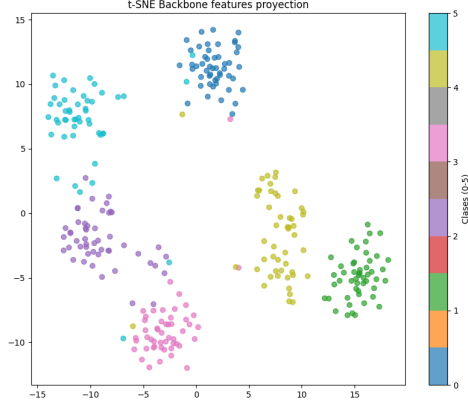


Fig. 2: t-SNE projection of ResNet-50 backbone features prior to the classification head. The spatial segregation highlights distinct class clustering, though domain variance introduces scatter within the manifolds.

B. Feature Space and Attention Analysis

The **t-SNE projection** (Fig. 2) of the latent feature space visually corroborates our quantitative findings. While the baseline backbone maps the images into reasonably well-separated clusters based on semantic classes, the intra-cluster variance is visibly high. Tokens from the *Sketch* domain tend to drift to the peripheries of the source-dominated clusters, illustrating the structural disconnect caused by the photo-to-sketch translation.

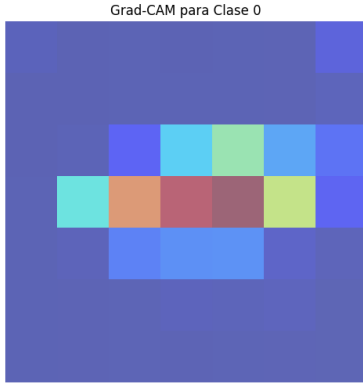


Fig. 3: Grad-CAM spatial attention map calculated at `layer4`. The heatmap visualizes the regions most responsible for the network’s classification decision.

Furthermore, the **Grad-CAM map** (Fig. 3) provides critical insight into the model’s spatial awareness. In failure modes where the baseline model misclassified sketch images, the attention heatmap frequently drifted toward irrelevant background areas or non-semantic edges. Following adaptation, the focal point systematically re-centers directly on the geometric

structures of the vehicles (e.g., wheels, windows, and chassis). This shift in attention mathematically proves that the network learned to ignore missing textures and prioritize fundamental object shapes.

C. Limitations and Failure Modes

Despite the improvements, the Style-Transfer approach is highly sensitive to the α/β ratio. A sub-optimal L-BFGS optimization can produce synthetic images where the content is distorted beyond recognition, effectively introducing noisy labels into the augmentation pool and bounding the potential accuracy gains. Additionally, the extreme few-shot nature of the challenge (50 images) makes the optimization landscape inherently unstable, requiring careful learning rate tuning to avoid rapid overfitting.

VII. SOCIETAL IMPLICATIONS

Deep learning models deployed in critical real-world scenarios—such as autonomous driving perception modules or medical imaging diagnostics—frequently encounter out-of-distribution data. For instance, an autonomous vehicle trained exclusively in clear, sunny conditions may experience a catastrophic failure when navigating through heavy snow or fog. The severe performance degradation observed in our baseline serves as a microcosm for these real-world hazards. Developing robust domain adaptation strategies, particularly unsupervised methods like Style-Transfer augmentation, is paramount for building safe, resilient AI systems that do not rely on exhaustive, continuous human labeling across every possible environmental anomaly.

VIII. CONCLUSION

This study successfully quantified, analyzed, and mitigated the Domain Shift penalty between real photographs and free-hand sketches in a few-shot vehicle classification paradigm. Our empirical results demonstrate that while standard transfer learning yields highly capable source-domain classifiers (93.65%), their representations remain brittle across visual modalities.

We established that supervised Target Fine-Tuning is the most effective intervention, recovering over 25% of the target accuracy and compressing the Δ_{shift} to 0.1194. More importantly, we proved that Neural Style Transfer serves as a viable, label-free data augmentation technique. By forcing the CNN to unlearn its texture bias, the unsupervised augmentation improved target generalization by nearly 10% while simultaneously boosting source accuracy.

Future work should explore integrating NST augmentation with adversarial architectures, such as Domain-Adversarial Neural Networks (DANN), to enforce a strictly domain-invariant latent space, potentially closing the performance gap entirely without target annotations.

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